

Circles and circle theorems - Set 1

TMUA practice questions by topic. Specification MM3.2, MM3.3, M5.8, M5.9. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Find the radius of the circle $x^2 + y^2 - 6x + 4y - 12 = 0$.
- A** 4
B 5
C 1
D $\sqrt{12}$
E $\sqrt{13}$
F 25
- 2 The circle with centre $(2, -1)$ passes through $(5, 3)$. Find the equation of the tangent to the circle at $(5, 3)$.
- A** $4x - 3y = 11$
B $3x + 4y = 27$
C $x + y = 8$
D $3x + 4y = 15$
E $3x - 4y = 3$
F $4x + 3y = 29$
- 3 The points $(1, 7)$ and $(5, -1)$ are the ends of a diameter of a circle. Find the equation of the circle.
- A** $(x - 3)^2 + (y - 3)^2 = 40$
B $(x - 2)^2 + (y - 4)^2 = 20$
C $(x - 3)^2 + (y - 3)^2 = \sqrt{20}$
D $(x - 3)^2 + (y - 3)^2 = 20$
E $(x - 3)^2 + (y - 3)^2 = 80$
F $(x + 3)^2 + (y + 3)^2 = 20$

- 4 A, B and C lie on a circle with centre O . Angle $AOB = 100^\circ$ and C lies on the major arc AB . Find angle ACB .
- A** 200°
B 100°
C 40°
D 50°
E 130°
F 80°
- 5 $PQRS$ is a cyclic quadrilateral with angle $P = 3x$, angle $R = x + 40^\circ$. Find x .
- A** 45°
B 50°
C 35°
D 30°
E 40°
F 25°
- 6 A chord of a circle of radius 13 is at perpendicular distance 5 from the centre. Find the length of the chord.
- A** 26
B $\sqrt{144}$
C 18
D 12
E 10
F 24

- 7 TA is a tangent to a circle at A , and AB is a chord. C is a point on the circle on the opposite side of AB from T , with angle $ACB = 65^\circ$. Find angle TAB .
- A** 25°
B 65°
C 115°
D 130°
E 50°
F 32.5°
- 8 The circle C has equation $x^2 + y^2 - 4x - 6y + 9 = 0$. Which of the following lines does not meet C ?
- A** $x = 4$
B $x = 2$
C $x = 0$
D $y = 0$
E $y = 5$
- 9 The circle $x^2 + y^2 = r^2$ and the line $3x + 4y = 20$ have exactly one point in common. Find r .
- A** $\frac{20}{7}$
B 4
C 20
D 5
E $\sqrt{20}$
F 2
- 10 AB is a diameter of a circle and C is a point on the circle with $AC = 6$ and $BC = 8$. Find the radius of the circle.
- A** $\sqrt{50}$
B 4
C 10
D 7
E 5
F 14

- 11** A circle passes through $(0, 0)$, $(6, 0)$ and $(0, 8)$. Find the coordinates of its centre.
- A** $(5, 5)$
 - B** $(2, 3)$
 - C** $(4, 3)$
 - D** $(6, 8)$
 - E** $(3, 4)$
 - F** $(3, 3)$
- 12** The circle C has centre (a, b) and radius r . P : ' $r > |b|$ '. Q : ' C crosses the x -axis at two distinct points'. Which one of the following is true?
- A** P is necessary and sufficient for Q .
 - B** P is necessary but not sufficient for Q .
 - C** P is sufficient but not necessary for Q .
 - D** P is neither necessary nor sufficient for Q .